"On my honor, I have neither given nor received any unauthorized and/or inappropriate assistance for all sessions of this exam. The work done on this exam is totally my own. I understand that by the school code, violation of these principles will lead to a zero grade and is subject to harsh discipline issues."

Soner Uyter
150719006

First, try to eliminate matrix A.

Question 1

$$A = \begin{bmatrix} 1 & \alpha & -\gamma & \alpha \\ 3\alpha & 5 & -9\gamma & 1 \\ 2 & \beta & 3 & \alpha^2 \\ 1 & 14 & -6 & 2 \end{bmatrix}$$

Some elim. matrices;

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & k & 0 & 1 \end{bmatrix} \quad \begin{bmatrix} 1 & 0 & 0 & 0 \\ -8 & 1 & 0 & 0 \\ M & 0 & 1 & 0 \\ -1 & 0 & 0 & 1 \end{bmatrix} \quad \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & -1/2 & 1 \end{bmatrix}$$

$\begin{matrix} -1 & 0 & 0 & 1 \\ -2 & 0 & 1 & 0 \end{matrix}$

$-R_1+R_4=R_4$
$-8R_1+R_2=R_2$
$M \cdot R_1+R_3=R_3$

Let $M=1$

$-R_1+R_4=R_4$ eliminasyonu bu matriste yer alıyor. (ilk elim. matrisi)

Let $M=1$

$$\begin{bmatrix} 1 & \alpha & -\gamma & \alpha \\ 3\alpha-8 & 5-8\alpha & -\gamma & 1-8\alpha \\ 2+M & 14+M\alpha & 3-M\gamma & \alpha^2+M\alpha \\ 0 & 14-\alpha & \gamma-6 & 2-\alpha \end{bmatrix} \xrightarrow{\text{Let } M=1} \begin{bmatrix} 1 & \alpha & -\gamma & \alpha \\ 3\alpha-8 & 5-8\alpha & -\gamma & 1-8\alpha \\ 3 & 14+\alpha & 3-\gamma & \alpha^2+\alpha \\ 0 & 14-\alpha & \gamma-6 & 2-\alpha \end{bmatrix} \xrightarrow{R_3+R_4=R_4} \begin{bmatrix} 1 & \alpha & -\gamma & \alpha \\ 3\alpha-8 & 5-8\alpha & -\gamma & 1-8\alpha \\ 3 & 14+\alpha & 3-\gamma & \alpha^2+\alpha \\ 0 & 28 & -3 & 2+\alpha^2 \end{bmatrix}$$

$-3R_1+R_3=R_3$

$$\begin{bmatrix} 1 & \alpha & -\gamma & \alpha \\ 3\alpha-8 & 5-8\alpha & -\gamma & 1-8\alpha \\ 0 & 14-2\alpha & 3+2\gamma & \alpha^2-2\alpha \\ 0 & 28 & -3 & 2+\alpha^2 \end{bmatrix} \xrightarrow{-\frac{R_3}{2}+R_4=R_4} \begin{bmatrix} 1 & \alpha & -\gamma & \alpha \\ 3\alpha-8 & 5-8\alpha & -\gamma & 1-8\alpha \\ 0 & 14-2\alpha & 3+2\gamma & \alpha^2-2\alpha \\ 0 & 35-\alpha & \frac{-3}{2}-\gamma & \frac{\alpha^2}{2}+2\alpha+2 \end{bmatrix}$$

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Question 2

k dim space. $x=5y$ x, y → axis The intersection with another geometry gives a volume.

$x=5y$

x, y → axis

$y=\frac{x}{5}$

In order to get a volume, we need 3-D space and 3 independent columns.

a) Minimum dimension of the space?

$x=5y$ $\quad x-5y=0$

$y=\frac{x}{5}$ } In this equation, the values x and y changes accordingly. It means that they are dependent columns.

In order to get a volume, we need at least 3-D space and 3 independent columns. So, k dimensional space must have minimum 4 dimensions. (Two of them independent, the other 2 are to get a volume)

b) We need minimum 4 dimensions (2 of them dependent)

$A \cdot x = b$ (general matrix form.)

x, y, z, t

for example $y=\frac{x}{5}$ 5, 10, 15, 20

$$\begin{bmatrix} 5 & 1 & k & l \\ 10 & 2 & m & n \\ 15 & 3 & 0 & p \\ 20 & 4 & r & s \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ t \end{bmatrix} = b$$

These columns are dependent.

These columns must be independent to get a volume. So;

$$\begin{bmatrix} k \\ m \\ 0 \\ r \end{bmatrix} \neq \alpha \cdot \begin{bmatrix} l \\ n \\ p \\ s \end{bmatrix}$$ (α is a constant)

c) $x+4y-z=6$

$x-5y=0$

$$\underbrace{\begin{bmatrix} 1 & 4 & -1 \\ 1 & -5 & 0 \end{bmatrix}}_{A\ 2\times 3} \underbrace{\begin{bmatrix} x \\ y \\ z \end{bmatrix}}_{x\ 3\times 1} = \underbrace{\begin{bmatrix} 6 \\ 0 \end{bmatrix}}_{b\ 2\times 1}$$

→ Yes, these vectors form a vector space.

Multiply both sides with A^T

$$\underbrace{\begin{bmatrix} 1 & 1 \\ 4 & -5 \\ -1 & 0 \end{bmatrix}}_{A^T\ 3\times 2} \underbrace{\begin{bmatrix} 1 & 4 & -1 \\ 1 & -5 & 0 \end{bmatrix}}_{A\ 2\times 3} \underbrace{\begin{bmatrix} x \\ y \\ z \end{bmatrix}}_{x} = \underbrace{\begin{bmatrix} 1 & 1 \\ 4 & -5 \\ -1 & 0 \end{bmatrix}}_{A^T} \underbrace{\begin{bmatrix} 6 \\ 0 \end{bmatrix}}_{b}$$

$$\begin{bmatrix} 2 & 1 & -1 \\ -1 & 41 & -4 \\ -1 & -4 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 6 \\ 41 \\ -6 \end{bmatrix}$$

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Question 2 part c cnt'd

$$\begin{bmatrix} 2 & 1 & -1 \\ -1 & 41 & -4 \\ -1 & -4 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 6 \\ 24 \\ -6 \end{bmatrix}$$

$$A \cdot x = b$$

$$\begin{aligned} 2x + y - z &= 6 \\ -x + 41y - 4z &= 24 \\ -x - 4y + z &= -6 \end{aligned}$$

$$x - 3y = 0$$

$$x = 3y$$

↓ dependent

$$\begin{aligned} 2x + y - z &= 6 \\ -x + 41y - 4z &= 24 \end{aligned}$$

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$U \cdot U^T$

$H^2 = I$

Question 4

First

$v(0) = 10L$

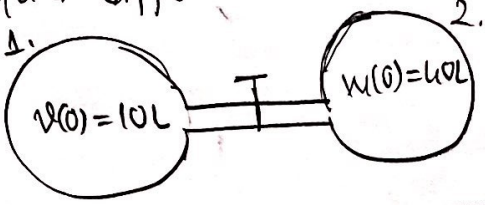
Sec

$w(0) = 40L$.

Liquid diffusion btw the tanks are proportional to the dif. v - w

$V_k \rightarrow$ Liquid diffused to the second tank, " " to the 1st tank

$W_k \rightarrow$



a) Soruya göre su tankları bu şekilde bağlı. Bu durumda musluk açıldığı takdirde aradaki su dengesi durağan değildir, sürekli bir su alış verişi gerçekleşir bu sebeple denklem continious olmalıdır.

	1.	2.	5dk sonra
	10L − 1L	+ 4L − 1L	+ 4L = 16
	40L − 4L	+ 1L − 4L	+ 1L = 34
	16L − 1	+ 4 − 1	+ 4 − 1 + 4 = 25
	34L − 4	+ 1 − 4	+ 1 − 4 + 1 = 23

2,

	10dk	1dk
1st	10L	1L
2nd	40L	4L

$$\frac{d}{dt}\begin{bmatrix} v \\ w \end{bmatrix} = \begin{bmatrix} 2 & -2 \\ -1 & 1 \end{bmatrix}\begin{bmatrix} v \\ w \end{bmatrix}$$

b) Y.'s of the mixture in tank 1 at time $t = 1$ sec.

Integrate both sides

$$\frac{dv}{dt}\begin{bmatrix} v \\ w \end{bmatrix} = \int \begin{bmatrix} 2 & -2 \\ -1 & 1 \end{bmatrix}\begin{bmatrix} v \\ w \end{bmatrix} = \int \begin{matrix} 2v - 2w \\ -v + w \end{matrix}$$

$= v^2 - w^2 + C_1$

$v(0) = 10L$

$w(0) = 40L$

$$\begin{bmatrix} v \\ w \end{bmatrix} = -\frac{v^2}{2} + \frac{w^2}{2} + C_2$$

c) for Steady state

$$V_{k+1} = V_k - x \cdot V_k + y \cdot W_k$$

$$W_{k+1} = W_k - y \cdot W_k + x \cdot V_k$$

$$\begin{bmatrix} V_{k+1} \\ W_{k+1} \end{bmatrix} = \begin{bmatrix} -x & +y \\ -y & +x \end{bmatrix}\begin{bmatrix} V_k \\ W_k \end{bmatrix}$$

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